VISHNUNARAYAN R NAGPAL | TE COMP 3 | ASSIGNMENT NO A10

In [1]:
```
pip install pandas matplotlib seaborn
```

```
Requirement already satisfied: pandas in /home/admin1/jupyter-env/lib/pyth
on3.12/site-packages (3.0.0)
Requirement already satisfied: matplotlib in /home/admin1/jupyter-env/lib/
python3.12/site-packages (3.10.8)
Requirement already satisfied: seaborn in /home/admin1/jupyter-env/lib/pyt
hon3.12/site-packages (0.13.2)
Requirement already satisfied: numpy>=1.26.0 in /home/admin1/jupyter-env/l
ib/python3.12/site-packages (from pandas) (2.4.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /home/admin1/jupy
ter-env/lib/python3.12/site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: contourpy>=1.0.1 in /home/admin1/jupyter-en
v/lib/python3.12/site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /home/admin1/jupyter-env/li
b/python3.12/site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /home/admin1/jupyter-e
nv/lib/python3.12/site-packages (from matplotlib) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /home/admin1/jupyter-e
nv/lib/python3.12/site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in /home/admin1/jupyter-en
v/lib/python3.12/site-packages (from matplotlib) (26.0)
Requirement already satisfied: pillow>=8 in /home/admin1/jupyter-env/lib/p
ython3.12/site-packages (from matplotlib) (12.1.0)
Requirement already satisfied: pyparsing>=3 in /home/admin1/jupyter-env/li
b/python3.12/site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: six>=1.5 in /home/admin1/jupyter-env/lib/py
thon3.12/site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

In [2]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Imports required libraries for data handling and data visualization.

In [3]:
```
df = pd.read_csv('iris.csv')
df.head()
```

Out[3]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

Loads the Iris dataset from a CSV file into a Pandas DataFrame for analysis.

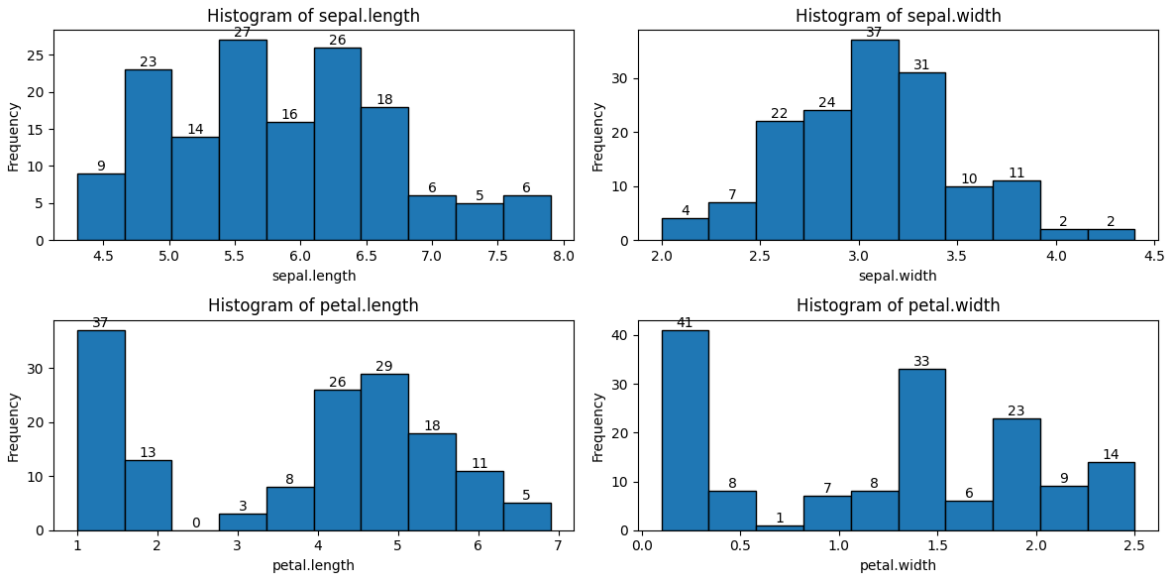
In [4]:
```
print("Feature and their types:")
df.dtypes
```

```
Feature and their types:
```

Out[4]:
```
sepal.length    float64
sepal.width     float64
petal.length    float64
petal.width     float64
variety             str
dtype: object
```

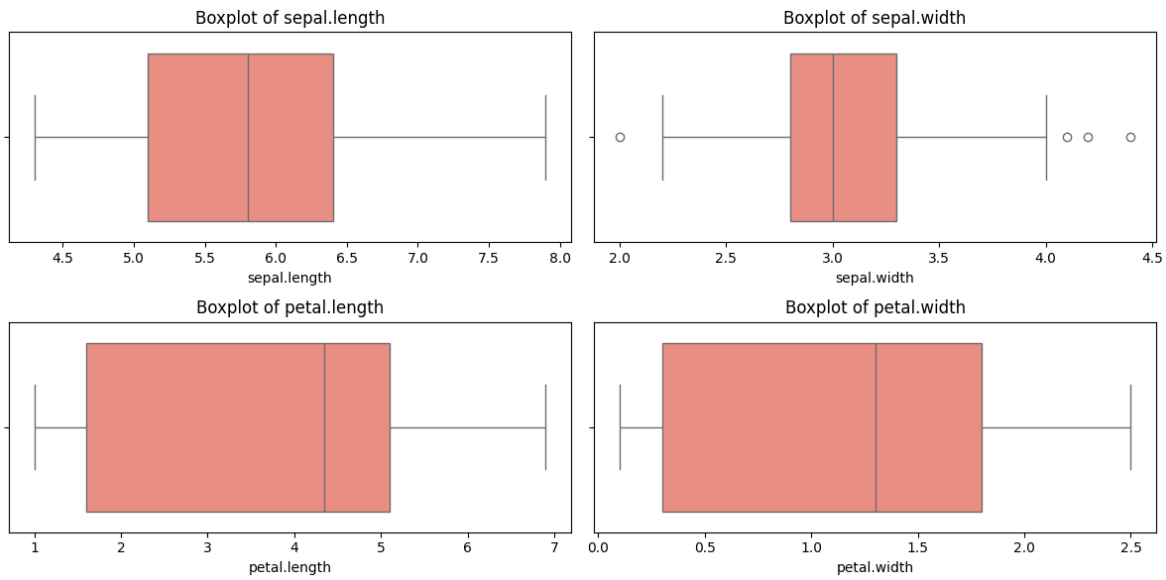
Displays all features along with their respective data types in the dataset.

In [5]:
```
plt.figure(figsize=(12, 6))

for i, column in enumerate(df.columns[:-1]):
    plt.subplot(2, 2, i + 1)

    ax = plt.hist(df[column], edgecolor="black")
    plt.gca().bar_label(plt.gca().containers[0], fmt='%d')
    plt.title(f"Histogram of {column}")
    plt.xlabel(column)
    plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```



Plots histograms to visualize the frequency distribution of each numerical feature.

In [6]:
```
plt.figure(figsize=(12, 6))
for i, column in enumerate(df.columns[:-1]):
    ax = plt.subplot(2, 2, i + 1)

    box_container = sns.boxplot(x=df[column], ax=ax, color='salmon')
    plt.title(f"Boxplot of {column}")

plt.tight_layout()
plt.show()
```



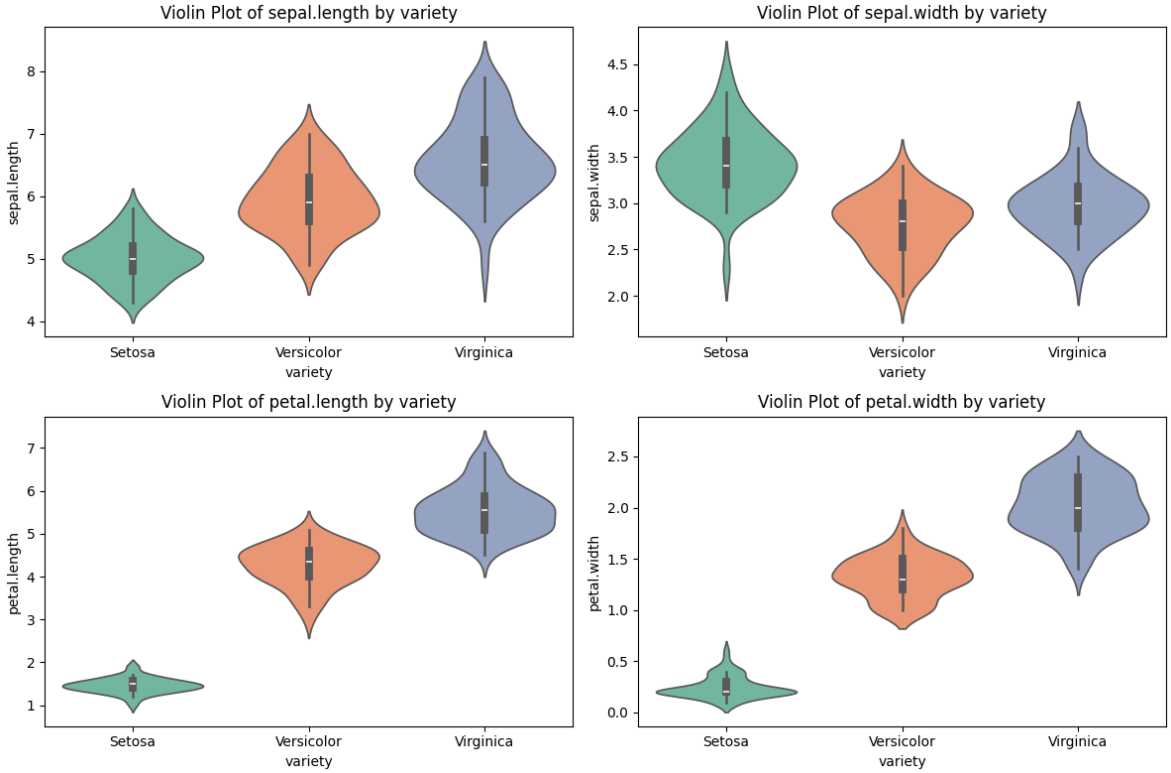
Uses boxplots to visualize data spread and identify potential outliers.

In [7]:
```
for column in df.columns[:-1]:
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)

    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = df[(df[column] < lower_bound) | (df[column] > upper_bound)

    print(f"\nFeature: {column}")
    print(f"  Mean: {df[column].mean():.2f}, Median: {df[column].median()
    print(f"  Outliers Detected: {'Yes' if not outliers.empty else 'No'}"
    print("-" * 40)
```

```
Feature: sepal.length
  Mean: 5.84, Median: 5.80, Std Dev: 0.83
  Outliers Detected: No
----------------------------------------

Feature: sepal.width
  Mean: 3.06, Median: 3.00, Std Dev: 0.44
  Outliers Detected: Yes
  Outlier Values: [4.4, 4.1, 4.2, 2.0]
----------------------------------------

Feature: petal.length
  Mean: 3.76, Median: 4.35, Std Dev: 1.77
  Outliers Detected: No
----------------------------------------

Feature: petal.width
  Mean: 1.20, Median: 1.30, Std Dev: 0.76
  Outliers Detected: No
----------------------------------------
```

Detects outliers using the IQR method and displays statistical measures for each feature.

In [8]:
```
plt.figure(figsize=(12, 8))
for i, column in enumerate(df.columns[:-1]):
    plt.subplot(2, 2, i + 1)
    sns.violinplot(x=df["variety"], y=df[column], palette="Set2", hue=df[
    plt.title(f"Violin Plot of {column} by variety")

plt.tight_layout()
plt.show()
```



Visualizes the distribution and density of features across different iris varieties.